

Nicola Novello

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Research Interests

Machine unlearning, Diffusion Models, classification with label noise, trustworthy ML.

Education

University of Klagenfurt <i>PhD in Information and Communication Engineering</i> Supervisors: Prof. Andrea M. Tonello, Prof. Deniz Gündüz	Oct. 2022 – Present
University of Padova <i>Master's Degree in Control Systems Engineering. Final Mark: 110/110 with honors</i> Supervisor: Prof. Alessandro Chiuso	Oct. 2020 – July 2022
University of Padova <i>Bachelor's Degree in Information Engineering. Final Mark: 110/110 with honors</i> Supervisor: Prof. Maria E. Valcher	Oct. 2017 – July 2020

Work Experience

University of Klagenfurt <i>Teaching Assistant</i> Taught the courses "Machine learning for information and communication engineering" and "Communication systems engineering"	Klagenfurt, Austria Mar. 2023 – Present
University of Klagenfurt <i>Project Assistant</i> Operated research for project "Power Line Communication for Metering - Data Analytics". In collaboration with Kelag International	Klagenfurt, Austria Apr. 2025 - Dec. 2025
Princeton University <i>Visiting Researcher</i> Performed research focused on neural channel capacity estimation. Under the supervision of Prof. H. Vincent Poor, department of electrical engineering	Princeton (NJ), US Jun. 2025 - Jul. 2025
University of Klagenfurt <i>Project Assistant</i> Conducted administrative duties for the European project ERASMUS+ NEON (Network of competence on IoT)	Klagenfurt, Austria Aug. 2022 - Feb. 2024
Trumpf <i>Student Intern</i> Developed deep learning algorithm in PyTorch for bending machine trajectory adjustment. Integrated the deep learning algorithm with existing C# infrastructure.	Vicenza, Italy Mar. 2022 - June 2022

Selected Publications

1. **Nicola Novello**, Federico Fontana, Luigi Cinque, Deniz Gundiz, and Andrea M. Tonello, "A Unified Framework for Diffusion Model Unlearning with f-Divergence" [🔗](#), ICML 2026
2. **Nicola Novello** and Andrea M. Tonello, "f-Divergence Based Classification: Beyond the Use of Cross-Entropy" [🔗](#), ICML 2024
3. **Nicola Novello** and Andrea M. Tonello, "Empty SPACE: Cross-Attention Sparsity for Concept Erasure in Diffusion Models" [🔗](#), arXiv preprint arXiv:2605.10198
4. **Nicola Novello** and Andrea M. Tonello, "Robust Classification with Noisy Labels Based on Posterior Maximization" [🔗](#), arXiv preprint arXiv:2504.06805

5. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Mutual Information Estimation via f -Divergence and Data Derangements" [🔗](#), NeurIPS 2024
6. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Copula Density Neural Estimation" [🔗](#), IEEE Transactions on Neural Networks and Deep Learning 2025

Other Publications

1. **Nicola Novello** and Andrea M. Tonello, "Data-Driven Grid Monitoring Under Label Noise Using Power Line Modems" [🔗](#), IEEE SmartGridComm 2025
2. **Nicola Novello** and Andrea M. Tonello, "Label Noise Resistant f -Divergence Receiver for Power Line Communications" [🔗](#), IEEE SmartGridComm 2024
3. **Nicola Novello** and Andrea M. Tonello, "Recurrent DQN for radio fingerprinting with constrained measurements collection" [🔗](#), ICT Express (2025)
4. Frank E. Ebong, **Nicola Novello**, and Andrea M. Tonello, "Human Detection Based on Learning and Classification of Radio Scattering Parameters and Para-Hermitian Eigenvalue Decomposition" [🔗](#), IEEE PIMRC 2024
5. **Nicola Novello** and Andrea M. Tonello, "Reinforcement learning for simplified training in fingerprinting radio localization" [🔗](#), IEEE BalkanCom 2023

Interests and Competences

Programming and software skills: Python (PyTorch, NumPy, Scikit-Learn), Matlab, C++, Java, L^AT_EX, GitHub

Expertise: Machine learning, signal processing, information theory

Languages: Italian (native), English (fluent)

Academic Service

Awarded "Gold Reviewer" at ICML 2026.

Reviewer for ICML, ICLR, NeurIPS, AAAI, IEEE Transactions on Communications, IEEE Transactions on Neural Networks and Learning Systems, IEEE VTC